

Weekly Homework 03

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Textbook Problems

#4 Suppose $x, y \in \mathbb{Z}$. If x and y are odd, then xy is odd.

Case 1: $x > 0$ and $y > 0$

Case 2: $x < 0$ and $y > 0$

Case 3: $x > 0$ and $y < 0$

Case 4: $x < 0$ and $y < 0$

Proof 1: Suppose $x = 2a + 1$, for all integers a , x is odd. W.L.O.G. we can say the same for y for $y = 2b + 1$.

$$\begin{aligned} xy &= (2a + 1)(2b + 1) = 4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1 \\ &\implies 2m + 1, \text{ which is odd.} \end{aligned}$$

#6 Suppose $a, b, c \in \mathbb{Z}$. If $a|b$ and $a|c$, then $a|(b + c)$.

Case 1: $b = 2m$ and $c = 2n$

Case 2: $b = 2m$ and $c = 2n + 1$

Case 3: $b = 2m + 1$ and $c = 2n + 1$

Proof 1: $b = a \times k$ and $c = a \times l$ should mean $b + c = a \times k + a \times l = a(k + l)$

$$2m + 2n = 2\left(\frac{ak}{2}\right) + 2\left(\frac{al}{2}\right) = 2\left(\frac{ak}{2} + \frac{al}{2}\right) = ak + al = a(k + l)$$

Proof 2: W.L.O.G.

$$\begin{aligned} 2m + 2n + 1 &= 2\left(\frac{ak}{2}\right) + 2\left(\frac{al - 1}{2} + 1\right) = 2\left(\frac{ak}{2} + \frac{al - 1}{2}\right) + 1 \\ &= ak + al - 1 + 1 = a(k + l) \end{aligned}$$

Proof 3: W.L.O.G.

$$\begin{aligned} 2m + 1 + 2n + 1 &= 2\left(\frac{ak - 1}{2}\right) + 1 + 2\left(\frac{al - 1}{2}\right) + 1 \\ &= 2\left(\frac{ak - 1}{2} + \frac{al - 1}{2}\right) + 2 = ak + al - 1 - 1 + 2 = a(k + l) \end{aligned}$$

#7 Suppose $a, b \in \mathbb{Z}$. If $a|b$, then $a^2|b^2$.

Case 1: If $a = 0$, then $b = 0$ and $a^2|b^2$.

Case 2: If $a \neq 0$, then $b = ak$ for some integer k . Therefore $b^2 = a^2k^2$ and $a^2|b^2$. W.L.O.G. applies here for all integers.

#14 If $n \in \mathbb{Z}$, then $5n^2 + 3n + 7$ is odd.

Case 1: $n = 2k$ for some integer k .

Case 2: $n = 2k + 1$ for some integer k .

Proof 1:

$$\begin{aligned} 5n^2 + 3n + 7 &= 5(2k)^2 + 3(2k) + 7 = 20k^2 + 6k + 7 \\ &= 20k^2 + 6k + 6 + 1 = 2(10k^2 + 3k + 3) + 1 \\ &\implies 2m + 1, \text{ which is odd.} \end{aligned}$$

Proof 2:

$$\begin{aligned} 5n^2 + 3n + 7 &= 5(2k + 1)^2 + 3(2k + 1) + 7 \\ &= 5(4k^2 + 4k + 1) + 6k + 3 + 7 = 20k^2 + 26k + 15 \\ &= 20k^2 + 26k + 14 + 1 = 2(10k^2 + 13k + 7) + 1 \\ &\implies 2m + 1, \text{ which is odd.} \end{aligned}$$

#16 If two integers have the same parity, then their sum is even.

Case 1: $k, l \in \mathbb{Z}$ and $k = 2m$ and $l = 2n$ for some integers m and n .

Case 2: $k, l \in \mathbb{Z}$ and $k = 2m + 1$ and $l = 2n + 1$ for some integers m and n .

Proof 1: $k + l = 2m + 2n = 2(m + n)$, which is even.

Proof 2: $k + l = 2m + 1 + 2n + 1 = 2(m + n + 1)$, which is even.

Additional Problems

1. Write problems in English sentences.

A. $\forall m \in \mathbb{Z}, \exists n \in \mathbb{Z}$ such that $m = 2n$.

For integers m there is an integer n such that $m = 2n$.

B. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $x = 2y$.

For real numbers x there is a real number y such that $x = 2y$.

2. An integer x is trodd if $x = 3a + 1$ for some $a \in \mathbb{Z}$

A.

a	$x = 3a + 1$
0	1
1	4
2	7
3	10
4	13

By this table, we can see that 2, 3, 5, 6, and 8 would not be trodd.

B. If x is a trodd integer, then x^2 is also a trodd integer.

$$\begin{aligned}\text{Proof: } x^2 &= (3a + 1)^2 = 9a^2 + 6a + 1 = 3(3a^2 + 2a) + 1 \\ &\implies x^2 \text{ is a trodd integer.}\end{aligned}$$

3. An integer x is four-six divisible if $4|x$ or $6|x$.

A. Prove: If x is four-six divisible, then $7x + 3$ is odd.

Case 1: If $4|x$, then $x = 4a$ for $a \in \mathbb{Z}$

Case 2: If $6|x$, then $x = 6b$ for $b \in \mathbb{Z}$

Proof 1: $7x + 3 = 7(4a) + 3 = 28a + 3 = 2(14a) + 3$, which is odd.

Proof 2: $7x + 3 = 7(6b) + 3 = 42b + 3 = 2(21b) + 3$, which is odd.